

Stage 1 Presentation Guidelines

The objective of this first presentation is to prove the rigor of your research. You are not just listing papers; you are pitching a cohesive, evolutionary story of your topic.

Please ensure your slide deck covers the following sections:

1. Topic Definition & The Core Problem

- **Topic:** Clearly state your assigned/chosen image processing task (e.g., Image Denoising, Edge Detection).
- **The Problem:** What is the overarching goal of this task? What were the inherent mathematical or visual challenges in solving it?

2. The Visual Evolutionary Timeline

- You must provide a clear, visual timeline mapping the evolution of algorithms for your topic.
- It must explicitly show the transitions: **Classical DIP** → **Early Deep Learning (e.g., CNNs)** → **Modern SOTA (e.g., Vision Transformers)**.
- **Bottlenecks:** Briefly indicate on the timeline *why* the field transitioned from one era to the next (What were the limitations of the previous era?).

3. Source Quota & 70/30 Distribution

You must present a master list of your **10+ distinct sources**, proving you meet the syllabus criteria:

- **Category A (Min. 70%):** Peer-reviewed academic papers.
 - *Must* be from top-tier venues (IEEE TPAMI, CVPR, ICCV, ECCV, NeurIPS, MICCAI).
 - *Must* include both older milestone papers and recent SOTA (2022–present).
- **Category B (Max. 30%):** Auxiliary materials (e.g., textbook chapters, Stanford/MIT lectures, Google/Meta technical blogs).

4. Paper-by-Paper Breakdown (The "Rigor" Check)

For the Category A academic papers on your list, We expect you to act as domain experts. For each paper, you must quickly highlight:

A. Metadata & Credibility:

- **Title & Authors:** Name of the paper and the leading researchers.
- **Institution:** What university or industry lab (e.g., MIT, Google DeepMind, Meta AI) are the writers from?
- **Venue & Year:** Where and when was it published? (Verify it is top-tier).
- **Citation Count:** How many citations does it currently have? (High citations validate it as a historical "milestone" paper).

B. Technical Specifications:

- **Topic/Sub-problem:** What specific bottleneck is this paper trying to solve?
- **Model/Architecture:** What is the name/structure of the model used? (e.g., Holistically-Nested Edge Detection, Swin Transformer).
- **Dataset(s):** What benchmark datasets were used for training/evaluation? (e.g., ImageNet, COCO).

C. Evolutionary Context (Crucial for Approval):

- **Overall Goal & Value:** What was the importance of this paper when it was published vs. right now? (Is it still used, or is it just a stepping stone?)
- **Relation to Other Papers:** You must connect the dots. How does this paper relate to the others on your list? (*Example: "This paper solved the spatial limitations of Paper 1, and introduced the attention mechanism that is perfected in Paper 3."*)

5. Category B Source Justification

For your auxiliary sources, briefly state what they are and how they will help you bridge the gap between academic theory and practical engineering for Stage 2 (Codebase Analysis).